

Syllabus:- Parabola
SECTION - I
Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The ends of a line segment are P (1, 3) and Q (1, 1). R is a point on the line segment PQ such that PR : QR = 1 : λ . If R is an interior point of the parabola $y^2 = Ax$, then
 (1) $\lambda \in (0, 1)$ (2) $\lambda \in \left(-\frac{3}{5}, 1\right)$ (3) $\lambda \in \left(\frac{1}{2}, \frac{3}{5}\right)$ (4) None of these
- The length of the common chord (in units) of the parabolas $y^2 = x$ and $x^2 = y$ is
 (1) $2\sqrt{2}$ (2) 1 (3) $\sqrt{2}$ (4) $\frac{1}{\sqrt{2}}$
- If point (2a, a) lies inside the parabola $x^2 - 2x - 4y + 3 = 0$ then interval of values of a is
 (1) $\left[\frac{1}{2}, \frac{3}{2}\right]$ (2) $\left(\frac{1}{2}, \frac{3}{2}\right)$ (3) (1, 3) (4) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$
- AB is a chord of the parabola $y^2 = 4ax$ with vertex at A. BC is drawn perpendicular to AB meeting the axis at C. The projection of BC on the x-axis is-
 (1) a (2) 2a (3) 4a (4) 8a
- Let L_1 be the length of the common chord of the curves $x^2 + y^2 = 9$ and $y^2 = 8x$, and L_2 be the length of the latus rectum of $y^2 = 8x$, then
 (1) $L_1 > L_2$ (2) $L_1 = L_2$ (3) $L_1 < L_2$ (4) $\frac{L_1}{L_2} = \sqrt{2}$
- A ray of light travels along the line $y = \frac{3}{2}$ and strikes the curve $y^2 = 2(x + y)$, then equation of reflected ray is
 (1) $4x + 3y - 3 = 0$ (2) $4x - 3y + 3 = 0$ (3) $4x + 3y + 3 = 0$ (4) $4x - 3y - 3 = 0$
- If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabolas $y^2 = 4ax$ and $x^2 = 4ay$, then –
 (1) $d^2 + (3b - 2c)^2 = 0$ (2) $d^2 + (3b + 2c)^2 = 0$ (3) $d^2 + (2b - 3c)^2 = 0$ (4) $d^2 + (2b + 3c)^2 = 0$
- A parabola has the origin as its focus and the line $x = 2$ as the directrix. Then the vertex of the parabola is at
 (1) (2, 0) (2) (0, 2) (3) (1, 0) (4) (0, 1)
- A circle is described whose center is the vertex and whose diameter is three-quarters of the latus rectum of a parabola $y^2 = 4ax$. The common chord of the circle and parabola is
 (1) $x = \frac{a}{2}$ (2) $x = -\frac{a}{2}$ (3) $x = \frac{a}{4}$ (4) None of these

10. Find the equation of the parabola, which has vertex (0, 0) and is symmetric about $y = -x$ and passes through the point (2, -3).
 (1) $3x^2 = -4y$ (2) $3x^2 = 4y$ (3) $2x^2 = -3y$ (4) $4x^2 = -3y^2$
11. If P (at^2 , $2at$) be one end of a focal chord of the parabola $y^2 = 4ax$, then the length of the chord is
 (1) $a\left(t - \frac{1}{t}\right)^2$ (2) $a\left(t - \frac{1}{t}\right)$ (3) $a\left(t + \frac{1}{t}\right)$ (4) $a\left(t + \frac{1}{t}\right)^2$
12. If the segment intercepted by the parabola $y^2 = 4ax$ with the line $lx + my + n = 0$ subtends a right angle at the vertex, then-
 (1) $4al + n = 0$ (2) $4al + 4am + n = 0$ (3) $4am + n = 0$ (4) $al + n = 0$
13. The coordinates of the focus of the parabola described parametrically by $x = 5t^2 + 2$, $y = 10t + 4$ are
 (1) (7, 4) (2) (3, 4) (3) (3, -4) (4) (-7, 4)
14. The Circle $x^2 + y^2 = 9$ meets the parabola $y^2 = 8x$ at P and Q. If the length of PQ is equal to ℓ then ℓ^2 is
 (1) 14 (2) 44 (3) 34 (4) 32
15. Find integral x- coordinates of the point on x-axis from which exactly three distinct chords of the circle $x^2 + y^2 = 25$ can be drawn which are bisected by the parabola $y^2 = 20x$.
 (1) 44 (2) 32 (3) 34 (4) 21
16. The equation of the latus rectum of a parabola is $x + y = 8$ and the equation of tangent at the vertex is $x + y = 12$ then the square of length of the latus rectum is
 (1) 444 (2) 332 (3) 344 (4) 128
17. Let O be the vertex and Q be any point on the parabola, $x^2 = 8y$. If the point P divides the line segment OQ internally in the ratio 1 : 3, then the locus of P is
 (1) $x^2 = 2y$ (2) $x^2 = y$ (3) $y^2 = x$ (4) $y^2 = 2x$
18. Angle between the tangents drawn to $y^2 = 4x$ at the points where it is intersected by the line $y = x - 1$ is π / k then the value of k is :
 (1) 4 (2) 3 (3) 1 (4) 2
19. Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the center C of the circle $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its center at P is
 (1) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$ (2) $x^2 + y^2 - 4x + 9y + 18 = 0$
 (3) $x^2 + y^2 - 4x + 8y + 12 = 0$ (4) $x^2 + y^2 - x + 4y - 12 = 0$
20. The focus of parabola $x^2 - 4x + 2y + 8 = 0$ is
 (1) $(3/2, -2)$ (2) $(5/2, -2)$ (3) $(2, -3/2)$ (4) $(2, -5/2)$

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. If normal chord at a point 't' on parabola $y^2 = 4x$ subtends a right angle at vertex then find the value of t^2
22. Let $x = 2t$, $y = \frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that $SA \perp BA$, where A is any point on the conic. If k is the ordinate of the centroid of $\triangle SAB$, then $18 \times \lim_{t \rightarrow 1}$ is equal to

23. Let PQ be a focal chord of the parabola $y^2 = 4x$. If the centre of a circle having PQ as its diameter lies on the line $\sqrt{5}y + 4 = 0$, then find the length of the chord PQ.
24. If the equation of the parabola, whose vertex is at (5, 4) and the directrix is $3x + y - 29 = 0$, $x^2 + ay^2 + bxy + cx + dy + k = 0$ then $a + b + c + d + k$ is equal to
25. If vertex of a parabola is (2, -1) and the equation of its directrix is $4x - 3y = 21$, then the length of its latus rectum is

ANSWER KEY_DPP-13

1.	(1)	2.	(3)	3.	(2)	4.	(3)	5.	(3)	6.	(1)	7.	(4)
8.	(3)	9.	(1)	10.	(1)	11.	(4)	12.	(1)	13.	(1)	14.	(4)
15.	(4)	16.	(4)	17.	(1)	18.	(4)	19.	(2)	20.	(4)	21.	(2)
22.	(13)	23.	(7.2)	24.	(-576)	25.	(8)						